

End Sem Exam

Nov 2025

Transport Phenomena in Materials Engineering

Course Code: MT-1-03 (TO)

Duration: 3 h

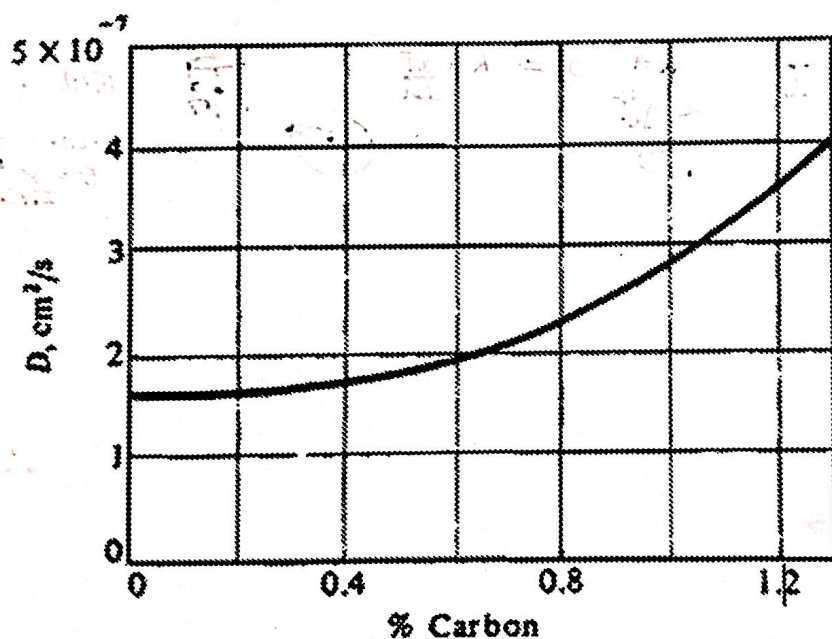
Materials Engineering, IIT Jammu

Max Marks: 50

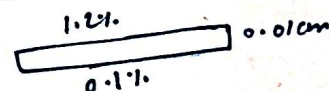
Q1) Explain the mechanism of mass transfer under substitutional diffusion. Derive the expression of diffusion coefficient for a binary diffusion couple made of Cu and Ni (3+7 Marks)

Q2) One side of an iron sheet, 0.01 cm thick, is subjected to a carburizing atmosphere at 1200 K such that a surface concentration of 1.2% carbon is maintained. The opposite face is maintained at 0.1% carbon. At steady state, determine the flux ($\text{mol cm}^{-2} \text{s}^{-1}$) of carbon through the sheet:

- a) if the diffusion coefficient is assumed to be independent of concentration ($D = 2 \times 10^{-7} \text{ cm}^2 \text{s}^{-1}$);
b) if the diffusion coefficient varies as per the curve/equation shown in the next page: (10 Marks)

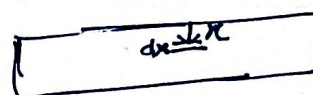


$$D(C) = \alpha + \beta C + \gamma C^2$$
$$D(C) = \alpha + \beta C + \gamma C^2$$
$$\alpha = 1.520 \times 10^{-7}$$
$$\beta = 5.50 \times 10^{-8}$$
$$\gamma = 1.25 \times 10^{-7}$$



(a) $J = -D \frac{\Delta C}{\Delta x}$

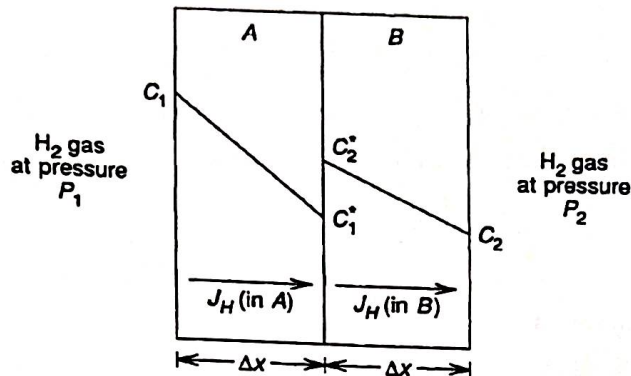
(b) $J =$



$$D \cdot \frac{\Delta C}{\Delta x}$$
$$(\alpha + \beta C + \gamma C^2) dC$$

$$dJ dx$$
$$\int_0^L J dx = \int_0^L D(C) \cdot dC$$
$$\int_0^L J dx = \int_0^L D(C) \cdot dC$$

Q3: Derive the expression of concentrations C_1^* and C_2^* for the hydrogen diffusing through a composite wall at steady state as shown below: (5 Marks)



Q4: Explain the difference between free and forced convection with suitable examples. (5 marks)

Q5: Derive the heat transfer equation for steady state in for the following geometries (5 marks)

(a) Hollow cylinder



(b) Hollow sphere



$$q_r = KA \frac{dT}{dr}$$



$$E/ds = \frac{q}{E}$$

$$E \cdot 2\pi r \frac{q}{E} = q$$

Q6: Explain the following terms with respect to fluid flow in brief: (2×5=10 marks)

a) Newton's Law of Viscosity

b) Reynold's Number

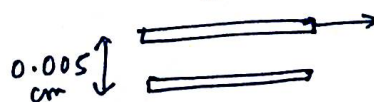
c) Couette Flow

d) Hagen-Poiseuille equation,

e) Friction losses in pipe flow

Q7: Two flat plates spaced 0.005 cm apart are separated by a lubricating oil of viscosity 0.2 Pa.s. If the lower plate is stationary and the upper plate moves with a velocity of 0.5 m/s, calculate the stress required to keep the upper plate in motion. (5 Marks)

$$\eta = 0.2 \text{ Pa.s}$$



$$v_n = \left(\frac{V}{y}\right)y$$

$$\tau_{xy} = -\frac{V}{y}\eta$$

$$\tau_{xy} = \frac{0.5 \text{ m/s}}{0.005 \times 10^{-2} \text{ m}} \times 0.2$$

$$\tau_{xy} = 2000 \text{ N/m}^2$$

$$\tau_{xy} = \eta \frac{\partial v}{\partial y}$$

$$\int \partial v = \frac{\eta}{V} \int \partial y$$

$$v = \frac{\eta}{V} y + C$$

$$5 \times 10^{-1} = \frac{-K}{\eta} \times 5 \times 10^{-3}$$

$$0.2 \times \frac{5 \times 10^{-1}}{5 \times 10^{-3}} = K$$

$$q = K A \frac{dT}{dx}$$